# Explainable Market Benchmarking and Price Decision-Support System for Short-Term Rentals

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Property Management SaaS Platform Project · Submission Deadline: May 7, 2026

*Abstract—Short-term rental hosts and property managers frequently set nightly prices without access to structured, data-driven market references. Informal competitor observation and intuition remain the dominant pricing practices, even among professional operators, leaving significant uncertainty around whether a given listing is competitively positioned within its local market. This paper presents the design and development of an Explainable Market Benchmarking and Price Decision-Support System for short-term rentals, built on observed Airbnb listing data across five major Italian cities: Rome, Milan, Florence, Venice, and Naples. The system combines a machine learning price estimation engine with a comparable-listings benchmarking workflow to generate market-aligned price ranges and classify each property as underpriced, aligned, or overpriced relative to similar listings in the same geographic micro-market and seasonal context. Critically, the system is designed not as a static pricing oracle, but as a transparent decision-support tool: feature-level explanations derived from interpretable machine learning techniques allow property managers to understand the specific drivers behind each price estimate and benchmark position, and to take informed, autonomous pricing actions accordingly. The proposed architecture separates the data pipeline, estimation logic, and user-facing dashboard into distinct layers, supporting both analytical reproducibility and practical deployment within a property management SaaS context. This paper constitutes Technical Draft #1 and presents the system design, dataset characterisation, and evaluation framework; empirical results will be reported in a subsequent draft once model training is complete.*

**Index Terms**—Short-term rentals, Airbnb pricing, explainable machine learning, market benchmarking, price estimation, comparable listings, decision support systems, geospatial micro-markets, SHAP values, gradient boosting.

## I. INTRODUCTION

The short-term rental (STR) market has expanded rapidly across major European tourist destinations, driven by the proliferation of peer-to-peer platforms such as Airbnb [1]. This growth has been accompanied by increasing professionalisation: property management companies and multi-listing hosts now constitute a substantial share of supply in most cities [2]. Despite this professionalisation, pricing decisions remain largely intuitive. Hosts and property managers typically rely on informal competitor observation, platform suggestions, or simple rule-of-thumb adjustments, without access to structured, transparent market evidence.

A critical gap exists between the richness of observable listing data and the tools available to practitioners. Raw data from platforms such as InsideAirbnb provides detailed snapshots of listing attributes and listed prices, yet no structured framework currently exists to translate this data into interpretable pricing intelligence for individual operators. The absence of such tools leads to persistent uncertainty in price setting and makes it difficult to justify pricing decisions in a consistent, auditable way.

This paper presents the design and development of an Explainable Market Benchmarking and Price Decision-Support System, applied to Airbnb listing data for five Italian cities: Rome, Milan, Florence, Venice, and Naples. Rather than proposing a black-box revenue management engine, the system is explicitly framed as a decision-support tool. Its purpose is to surface structured market evidence, provide interpretable price estimates, identify comparable listings, and communicate the drivers of each estimate in a way that empowers property managers to take informed, autonomous pricing actions.

The system is proposed around four functional modules: (i) geographic micro-market construction using neighbourhood boundaries as baseline and density-based spatial clustering as a validated refinement; (ii) price estimation using interpretable and tree-based machine learning models; (iii) benchmark range construction using quantile regression and comparable-listing retrieval; and (iv) an explainability layer based on SHAP values and what-if analysis. This modular architecture allows each component to be evaluated independently while contributing to a unified decision-support output.

The remainder of this paper is organised as follows. Section II reviews the relevant literature on STR pricing, hedonic price models, machine learning for price prediction, and explainable AI. Section III describes the dataset characteristics and preliminary target analysis. Section IV presents the methodology, including candidate model analysis, evaluation protocol, and benchmark inference logic. Section V outlines the evaluation framework. Section VI concludes with a discussion of scope limitations and future work.

## II. LITERATURE REVIEW

### A. STR Pricing and the Airbnb Market

The short-term rental market has attracted substantial academic attention since Airbnb's rapid global expansion. Zervas et al. [3] documented the platform's disruptive impact on traditional hospitality, while Guttentag [1] characterised it as an informal accommodation sector operating outside conventional regulatory frameworks. Studies on STR pricing have consistently identified location, property size, listing type, and host reputation as primary price determinants [2][4]. In the European context, Gyodi and Nawaro [5] analysed Airbnb prices across ten major EU cities and confirmed that attributes related to size, quality, and location are all significant drivers of nightly rates, with location effects showing strong spatial dependence across all cities.

At the city level, several studies have documented the heterogeneity of pricing structures across neighbourhoods. Wang and Nicolau [4] analysed listings from 33 cities on Airbnb and found that room type, size, and guest reviews systematically explain price variation. Deboosere et al. [6] further showed that access to public transportation and neighbourhood socioeconomic factors have significant effects in New York. The Italian STR market has received comparatively less attention, motivating the geographic scope of the present study.

### B. Hedonic Price Models

The theoretical foundation of STR price modelling lies in the hedonic price framework, originally formalised by Rosen [7] building on Lancaster's characteristics theory [8]. Under this framework, the price of a product reflects the consumer's willingness to pay for each of its observable attributes. In the hospitality literature, hedonic models have been widely used to decompose hotel prices into structural, locational, and qualitative components [9][10]. Applied to Airbnb, the same logic implies that a listing's nightly price is a function of its physical characteristics, location premium, host reputation signals, and seasonal context.

A key limitation of standard hedonic OLS models in this domain is their assumption of spatial independence. Gyodi and Nawaro [5] demonstrated that Airbnb prices exhibit significant spatial autocorrelation in all ten European cities they analysed, and that OLS estimates are biased, particularly for location variables. Islam et al. [11] extended this line of inquiry by incorporating Moran Eigenvector Spatial Filtering into an XGBoost model, showing that explicitly addressing spatial dependence through eigenvector decomposition substantially improves prediction accuracy. These findings motivate careful attention to spatial structure in micro-market construction, and suggest that data-driven segmentation may offer improvements over administrative boundaries in markets with strong spatial heterogeneity—a hypothesis we propose to test empirically.

### C. Machine Learning for Price Prediction

Beyond classical regression, machine learning approaches have demonstrated strong performance for tabular price prediction tasks. Islam et al. [11] showed that XGBoost outperforms random forests and linear regression models on Airbnb data when spatial information is incorporated. CatBoost [12] and LightGBM [13] have demonstrated consistent advantages over random forests on large tabular datasets with high-cardinality categorical features, making them particularly suitable for STR data where city, neighbourhood, and room type are key categorical predictors.

Generalised Additive Models (GAMs) [14] represent an intermediate approach: they capture non-linear relationships between continuous features and price while maintaining the interpretability of component-wise effect curves. This property is especially relevant for spatial features such as distance to city centre, latitude, and longitude, where the price premium is known to be non-linear [5]. Regularised linear models such as Ridge and Elastic Net serve as essential baselines, providing interpretable coefficient estimates and enabling quantification of the incremental value added by more complex approaches.

### D. Explainable Artificial Intelligence (XAI)

The shift toward machine learning in pricing applications raises a fundamental tension between predictive accuracy and interpretability. In decision-support contexts—where a human operator must understand, trust, and act on model outputs—black-box predictions alone are insufficient. Lundberg and Lee [15] introduced SHAP (SHapley Additive exPlanations) as a unified framework for model-agnostic local explanations, grounded in cooperative game theory. SHAP values assign each feature a contribution to a given prediction, decomposing the output in a way that is both locally accurate and globally consistent. Molnar [16] provides a comprehensive treatment of the broader interpretable ML landscape within which SHAP sits.

In the STR context, SHAP-based explanations allow property managers to understand not only the estimated market price, but why that price is what it is: which features contribute positively or negatively, and by how much. Extending this to counterfactual or what-if analysis—asking how the estimate would change if the property were 500 metres closer to the city centre, or had one additional bedroom—transforms the system from a static estimator into a prescriptive decision-support tool. This design philosophy distinguishes our approach from existing dynamic pricing solutions, which typically optimise revenue without communicating the reasoning behind their recommendations.

## III. DATASET AND PRELIMINARY TARGET ANALYSIS

### A. Data Source and Description

The dataset consists of Airbnb listing snapshots sourced from InsideAirbnb for five Italian cities: Rome, Milan, Florence, Venice, and Naples. The data comprises four seasonal snapshots corresponding to Early Spring, Early Summer, Early Autumn, and Early Winter, yielding a total of 282,047 listing-level observations after preprocessing and quality filtering. Each observation includes structural attributes (room type, accommodation capacity, number of bedrooms and bathrooms), host characteristics (superhost status, number of listings), review scores (overall rating, cleanliness, communication), geographic coordinates, and the observed nightly price in EUR.

An important limitation of this dataset is that observed prices reflect listed prices rather than transacted prices. Furthermore, the snapshot structure prevents the construction of continuous booking histories or occupancy series. The system is therefore explicitly scoped as a market benchmarking and pricing intelligence tool, not a revenue forecasting or dynamic pricing engine.

### *B. Preliminary Target Analysis*

A preliminary descriptive analysis of the target variable reveals a strongly right-skewed nightly price distribution. In the full listing-level dataset (n = 282,047), the mean nightly price is EUR 173.16, while the median is EUR 120.00, with Q1 = EUR 85 and Q3 = EUR 174. The 95th percentile is EUR 380 and the 99th percentile is EUR 900, whereas the maximum observed value reaches EUR 95,195. This confirms the presence of substantial outliers and motivates the use of robust descriptive summaries, log-price transformation during model training, and complementary evaluation metrics.

Price distributions differ substantially across cities. Median nightly prices are highest in Venice (EUR 153), followed by Florence (EUR 136), Rome (EUR 127), Milan (EUR 107), and Naples (EUR 86). This cross-city heterogeneity motivates city-aware model specifications and local benchmarking rather than a single global model. Seasonal variation is also evident: median prices range from EUR 131 in Early Autumn to EUR 103 in Early Winter, confirming that seasonal context must be incorporated into both the estimation engine and the benchmark construction workflow.

These findings carry direct methodological implications: (i) the target must be log-transformed during model training to reduce the influence of extreme values; (ii) evaluation metrics must include robust alternatives to RMSE; and (iii) the benchmarking module must operate at the micro-market level rather than at city-wide aggregation to be practically useful.

## IV. METHODOLOGY

### *A. System Architecture*

The system is designed as a pipeline of four functional modules, each addressing a distinct analytical subtask. The modularity of the architecture allows each component to be evaluated independently while contributing to a unified decision-support output. The four modules are: (1) geographic micro-market segmentation; (2) price estimation; (3) benchmark range construction and comparable listing retrieval; and (4) the explainability layer.

### *B. Module 1 — Geographic Micro-Market Construction*

The proposed micro-market construction follows a two-stage strategy. In the first stage, official neighbourhood boundaries provided by municipal sources serve as the baseline segmentation. Administrative neighbourhoods are interpretable, stable across snapshots, and directly communicable to property managers. They represent the defensible default if no data-driven refinement proves beneficial.

In the second stage, density-based spatial clustering—specifically HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) [17]—is evaluated as a refinement of the baseline. HDBSCAN is a candidate because it does not require the number of clusters to be specified a priori, it is robust to spatially non-uniform density distributions characteristic of Italian STR markets, and it can classify low-density peripheral listings as noise rather than forcing them into poorly-defined clusters. The clustering refinement will be adopted only if it demonstrably improves within-segment price homogeneity, reduces model MAE, and produces stable boundaries across seasonal snapshots. If these criteria are not met, the administrative baseline is retained. This conditional adoption strategy ensures that micro-market construction remains both methodologically rigorous and practically defensible to industry stakeholders.

### *C. Module 2 — Price Estimation Engine*

The estimation engine adopts a two-level strategy. A regularised linear baseline (Ridge or Elastic Net) provides an interpretable reference model with directly readable coefficients. Ridge is preferred when features are highly correlated (e.g., accommodates and bedrooms), as it distributes rather than eliminates correlated predictors. Elastic Net is considered as an alternative if sparse feature selection is desirable. These baselines serve to confirm that the data contains genuine signal and to quantify the incremental value of more complex models.

The primary estimator is either CatBoost or a Generalised Additive Model (GAM). CatBoost is a strong candidate given the dataset's large size (282,047 observations), high proportion of categorical features, and known interactions between city, neighbourhood, room type, and capacity. It handles categorical variables natively without requiring encoding, reducing preprocessing complexity. Its principal failure risk is overfitting in sparse micro-markets with few listings. A GAM is considered as an alternative primary model: it captures non-linear relationships between continuous features (distance to centre, latitude, longitude, review scores) and price while maintaining full interpretability through visualisable component curves. If the professor's evaluation criteria emphasise interpretability over predictive accuracy, the GAM may be preferred. LightGBM or XGBoost serve as a second strong estimator for empirical comparison. All models are trained on log-transformed prices.

### *D. Module 3 — Benchmark Range and Comparable Listings*

Estimating a point price is insufficient for practical decision-support: operators need to know whether their observed price falls within the plausible range for their market segment. Quantile regression (or quantile boosting) is used to estimate the 10th and 90th conditional percentiles of the price distribution, producing a benchmark band rather than a single estimate. This band defines the operational classification: listings priced below the lower bound are underpriced; those above the upper bound are overpriced; the remainder are aligned.

Comparable listing retrieval uses a k-Nearest Neighbours similarity engine operating in the feature space of micro-market identity, room type, accommodation capacity, and selected quality indicators. The returned comparables are actual listings from the dataset, providing human-readable reference points that anchor the benchmark to observable market reality. kNN is not used as a primary price predictor due to the curse of dimensionality and sensitivity to feature scaling; its role is restricted to retrieval.

### *E. Module 4 — Explainability Layer*

SHAP values [15] are computed for every prediction produced by the primary estimator. SHAP is model-agnostic and provides locally accurate, globally consistent explanations regardless of the underlying model. For each listing, the explainability layer will communicate: (i) which features are driving the estimated price up or down; (ii) the magnitude of each feature's contribution in EUR (back-transformed from log scale); and (iii) a what-if analysis showing how the estimate would change under hypothetical attribute modifications. A potential failure mode is the unintuitive splitting of contribution between correlated features (e.g., accommodates and bedrooms); this will be addressed through feature grouping in the presentation layer.

### *F. Benchmark Inference Logic*

The final benchmark output integrates evidence from multiple modules rather than relying on a single estimator. The benchmark centre is a weighted combination of the global model prediction and the local comparable median, giving greater weight to the comparable median when the micro-market contains sufficient observations and to the global model when comparables are sparse. The benchmark range is derived from the quantile model outputs, optionally widened by the dispersion of comparable listing prices. This hybrid inference logic is more interpretable and defensible than a classical ensemble of regressors, and avoids the opacity that would result from averaging many models with unclear individual roles.

### *G. Evaluation Protocol and Metrics*

Models will be evaluated on a held-out test set constructed via a two-stage grouped and stratified split. In the first stage, all snapshots belonging to the same listing_id are assigned to a single partition, ensuring that no listing appears in both training and test sets across seasonal observations. This grouped assignment is applied before any stratification to eliminate the primary leakage risk in the dataset. In the second stage, the grouped listing-level units are split into train, validation, and test partitions (70% / 15% / 15%) using stratified sampling on the joint city-by-season stratum, so that each city and each seasonal snapshot is proportionally represented in all three partitions. The implementation uses scikit-learn's GroupShuffleSplit for the grouping step followed by StratifiedShuffleSplit on the resulting groups. This two-stage procedure is necessary because a single stratified random split would not respect listing identity across snapshots, leading to optimistically biased error estimates.

Primary evaluation metrics are Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), both computed on back-transformed (EUR) predictions. MAE is selected for its direct monetary interpretability; RMSE complements it by assigning greater penalty to large errors, which is important given the right-skewed target distribution. Median Absolute Error (MedAE) is reported as a robustness-oriented secondary metric. $R^2$ is reported optionally as an indicator of overall variance explained.

For the benchmarking module, two additional metrics are planned: *benchmark interval coverage* (the proportion of listings whose observed price falls within the predicted benchmark range) and *average benchmark width* (the mean EUR width of the range). Pinball loss will be used to evaluate quantile regression calibration directly.

Given that explainability is central to the system's purpose and is explicitly named in the title, a dedicated explanation quality evaluation framework is also proposed. Three dimensions will be assessed. *Fidelity* measures how accurately the SHAP explanation approximates the model's actual behaviour at the local level; it will be evaluated by comparing SHAP-reconstructed predictions against true model outputs on held-out examples. *Stability* measures whether similar listings receive similar explanations; it will be operationalised as the rank correlation of SHAP feature importances across nearest-neighbour listing pairs. *User-perceived usefulness* will be assessed qualitatively during the industry stakeholder review session (G5 in the project timeline), using structured feedback questions on whether the explanations are actionable and comprehensible to non-technical property managers. Together, these three dimensions provide a more complete evaluation of the explainability layer than predictive metrics alone.

## V. EVALUATION PLAN

As this paper constitutes Technical Draft #1, the system is at the design stage and no empirical model results are yet available. This section defines the validation plan that will be executed in subsequent phases of the project, specifying the criteria against which each module will be assessed.

For the price estimation module, the primary success criterion is that the primary estimator (CatBoost, LightGBM, or GAM) achieves a meaningfully lower MAE than the regularised linear baseline on the held-out test set, confirming that non-linear and interaction effects add predictive value beyond the hedonic baseline. The magnitude of this improvement will determine whether the added complexity is justified for the decision-support use case.

For the micro-market module, the clustering refinement will be retained only if three criteria are jointly satisfied on the validation set: (i) within-segment price variance is lower under clustering than under administrative boundaries; (ii) model MAE improves when micro-market cluster identity replaces administrative neighbourhood as a feature; and (iii) cluster assignments are stable across at least three of the four seasonal snapshots, measured by adjusted Rand index.

For the benchmarking module, the target coverage rate for the 10th-90th percentile interval is at least 70% empirically, with an average benchmark width that is operationally informative—wide enough to be reliably calibrated but narrow enough to distinguish underpriced, aligned, and overpriced listings in each micro-market. For the explainability layer, the evaluation criteria are those described in Section IV-G: fidelity, stability, and stakeholder usefulness. All results will be reported in Technical Draft #2.

## VI. CONCLUSION

This paper has presented the design of an Explainable Market Benchmarking and Price Decision-Support System for short-term rentals, applied to Airbnb listing data across five Italian cities. The system's key contribution is its framing as a transparent decision-support tool rather than a black-box pricing engine: by combining interpretable price estimation, quantile-based benchmark ranges, comparable-listing retrieval, and SHAP-based explanations, the system provides property managers with structured market evidence and the reasoning behind each price estimate.

The preliminary target analysis confirms that the dataset contains rich pricing signal structured by geography, room type, and seasonality, while also revealing the distributional challenges (strong right skew, cross-city heterogeneity) that motivate the methodological choices described in Section IV. Empirical validation will be reported in Technical Draft #2 once model training is complete.

Future extensions, contingent on richer data availability, could include occupancy-related modelling, price-demand elasticity analysis, and integration of event-level demand signals. These extensions are outside the present scope, which is intentionally limited to market-aligned pricing intelligence based on observed listing data.

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